Markov Chains (AI HL)

ANSWER KEY

Section 1: Transition Matrices and State Vectors

1 In a standard transition probability matrix T (where entry T_{ij} represents the probability of moving from state i to state j), what must the sum of the entries in each row equal?

a. 1 ✓

b. 0

c. The number of states

d. $\frac{1}{n}$

Because each row corresponds to the probability distribution of transitioning from a given state to all possible next states, the sum of probabilities across each row is strictly 1.

2 Given initial state probability vector $\mathbf{s}_0 = [0.6, 0.4]$ and transition matrix $T = [[0.8, 0.2], [0.3, 0.7]]$, what is the state distribution after one step $\mathbf{s}_1 = \mathbf{s}_0 T$?

a. $[0.60, 0.40]$ ✓

b. $[0.55, 0.45]$

c. $[0.66, 0.34]$

d. $[0.48, 0.28]$

$\mathbf{s}_1 = [0.6(0.8) + 0.4(0.3), 0.6(0.2) + 0.4(0.7)] = [0.48 + 0.12, 0.12 + 0.28] = [0.60, 0.40]$.

3 What is the Markov property (memoryless property) in stochastic processes?

a. **The future state depends only upon the current state, not on the sequence of states that preceded it** ✓

b. The process has zero variance

c. All states have equal probability of occurrence

d. The chain can never visit the same state twice

A first-order Markov chain satisfies $P(X_{n+1} = s \mid X_n, X_{n-1}, \ldots, X_0) = P(X_{n+1} = s \mid X_n)$, meaning the conditional probability distribution depends only on the present state.

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Section 2: Multi-Step Transitions

4 How is the probability distribution $\mathbf{s}_n$ after n transitions computed from the initial state vector $\mathbf{s}_0$ and transition matrix T?

a. $\mathbf{s}_n = \mathbf{s}_0 T^n$ ✓

b. $\mathbf{s}_n = \mathbf{s}_0 + nT$

c. $\mathbf{s}_n = T^n \mathbf{s}_0$

d. $\mathbf{s}_n = (\mathbf{s}_0 T)^n$

Iterating the transition step $\mathbf{s}_1 = \mathbf{s}_0 T$, $\mathbf{s}_2 = \mathbf{s}_1 T = \mathbf{s}_0 T^2$ leads to $\mathbf{s}_n = \mathbf{s}_0 T^n$.

5 If $T = [[0.5, 0.5], [0.5, 0.5]]$, evaluate T^2.

a. $[[0.5, 0.5], [0.5, 0.5]]$ ✓

b. $[[0.25, 0.25], [0.25, 0.25]]$

c. $[[1, 0], [0, 1]]$

d. $[[0.5, 0], [0, 0.5]]$

Each entry is $0.5(0.5) + 0.5(0.5) = 0.25 + 0.25 = 0.5$. Thus $T^2 = T$.

6 What does entry $(T^3)_{2,1}$ represent?

a. **The probability of transitioning from State 2 to State 1 in exactly 3 steps** ✓

b. The probability of transitioning from State 1 to State 2 in 3 steps

c. Three times the one-step probability

d. The cube of the probability from State 2 to State 1

In the n-step transition matrix T^n, the entry in row i, column j is the conditional probability of being in state j after n steps given starting in state i.

Section 3: Steady-State (Stationary) Distribution

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7 What defining matrix equation must the steady-state probability vector $\boldsymbol{\pi} = [\pi_1, \pi_2]$ satisfy?

a. $\boldsymbol{\pi}T = \boldsymbol{\pi}$ **and** $\sum \pi_i = 1$ ✓

b. $T\boldsymbol{\pi} = \mathbf{0}$

c. $\boldsymbol{\pi}T = \mathbf{0}$

d. $\boldsymbol{\pi} = T^{-1}$

A steady-state distribution is stationary under transitions ($\boldsymbol{\pi}T = \boldsymbol{\pi}$) and its components form a valid probability distribution (sum to 1).

8 Find the stationary distribution for $T = [[0.7, 0.3], [0.2, 0.8]]$.

a. $\boldsymbol{\pi} = [0.4, 0.6]$ ✓

b. $\boldsymbol{\pi} = [0.5, 0.5]$

c. $\boldsymbol{\pi} = [0.3, 0.7]$

d. $\boldsymbol{\pi} = [0.6, 0.4]$

Set $[\pi_1, \pi_2][[0.7, 0.3], [0.2, 0.8]] = [\pi_1, \pi_2] \Rightarrow 0.7\pi_1 + 0.2\pi_2 = \pi_1 \Rightarrow 0.2\pi_2 = 0.3\pi_1 \Rightarrow 3\pi_1 = 2\pi_2$. Using $\pi_1 + \pi_2 = 1$: $\pi_1 + 1.5\pi_1 = 1 \Rightarrow 2.5\pi_1 = 1 \Rightarrow \pi_1 = 0.4$, $\pi_2 = 0.6$.

9 When is a Markov chain guaranteed to have a unique stationary distribution that is independent of the initial state?

a. **When the transition matrix is regular (some power T^k has strictly positive entries)** ✓

b. Only if the matrix is symmetric

c. Whenever $\det(T) = 0$

d. Only if there are 2 states

By the Perron-Frobenius theorem, a regular Markov chain has a unique stationary distribution, and $\lim_{n\to\infty} \mathbf{s}_0 T^n = \boldsymbol{\pi}$ for any initial distribution $\mathbf{s}_0$.

Section 4: Absorbing Markov Chains

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10 What is an absorbing state in a Markov chain?

a. **A state that, once entered, cannot be left ($T_{ii} = 1$ and $T_{ij} = 0$ for $j \neq i$)** ✓

b. A state that has probability zero of being visited

c. A state that transitions with equal probability to all other states

d. A state where the steady-state probability is 0

An absorbing state has a loop transition probability of 1 to itself, trapping the process indefinitely once entered.

11 In an absorbing Markov chain in canonical form $P = [[I, 0], [R, Q]]$, what does the fundamental matrix $N = (I - Q)^{-1}$ calculate?

a. **The expected number of visits to each transient state before absorption** ✓

b. The probability of absorption in one step

c. The eigenvalues of the transition matrix

d. The total number of absorbing states

The fundamental matrix $N = (I - Q)^{-1}$ provides the expected time (number of steps) spent in each transient state starting from any transient state before reaching an absorbing state.

12 How is the absorption probability matrix B calculated from fundamental matrix N and transient-to-absorbing matrix R?

a. $B = NR$ ✓

b. $B = RN$

c. $B = N + R$

d. $B = (I - Q)R$

The matrix of absorption probabilities from each transient state into each absorbing state is given by the matrix product $B = NR$.

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